§ 3 基本应用一

§ 3.1 一维谐振子

$$H = \frac{p^2}{2m} + \frac{1}{2} m\omega^2 q^2$$

$\Big\downarrow$ 正则量子化

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{q}^2 , \quad [\hat{q}, \hat{p}] = i\hbar$$

坐标表象下定态 Schrodinger方程，

$$\left(-\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2} m\omega^2 x^2\right)\psi(x) = E\psi(x)$$

$$\hat{a} = \frac{1}{\sqrt{2m\hbar\omega}}(i\hat{p} + m\omega\hat{q})$$

$$\hat{a}^{\dagger} = \frac{1}{\sqrt{2m\hbar\omega}}(-i\hat{p} + m\omega\hat{q})$$

$$[\hat{a}, \hat{a}^{\dagger}] = \frac{1}{2m\hbar\omega} 2im\omega [\hat{p}, \hat{q}] = \frac{-2im\omega i\hbar}{2m\hbar\omega} = 1 \Rightarrow \hat{a}\hat{a}^{\dagger} = \hat{a}^{\dagger}\hat{a} + 1$$

$$\Rightarrow \quad \hat{p} = \frac{\sqrt{2m\hbar\omega}}{2i}(\hat{a} - \hat{a}^{\dagger}) = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^{\dagger} - \hat{a}) \qquad \hat{p} = i\sqrt{\frac{m\hbar\omega}{2}}(\hat{a}^{\dagger} - \hat{a})$$

$$\hat{q} = \frac{\sqrt{2m\hbar\omega}}{2m\omega}(\hat{a} + \hat{a}^{\dagger}) = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^{\dagger} + \hat{a}) \qquad \hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^{\dagger} + \hat{a})$$

$$\begin{aligned}
\hat{H} &= \frac{1}{2m}(-1)\frac{m\hbar\omega}{2}(\hat{a}^{\dagger} - \hat{a})^2 + \frac{1}{2}m\omega^2\frac{\hbar}{2m\omega}(\hat{a}^{\dagger} + \hat{a})^2 \\
&= -\frac{1}{4}\hbar\omega(\hat{a}^{\dagger} - \hat{a})^2 + \frac{1}{4}\hbar\omega(\hat{a}^{\dagger} + \hat{a})^2 \\
&= \frac{1}{2}\hbar\omega(\hat{a}^{\dagger}\hat{a} + \hat{a}\hat{a}^{\dagger}) \qquad \hat{a}\hat{a}^{\dagger} = \hat{a}^{\dagger}\hat{a} + 1 \\
&= \hbar\omega\left(\hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right)
\end{aligned}$$

$$[\hat{a}, \hat{H}]$$

$$= \hbar\omega\left[\hat{a}, \hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right] = \hbar\omega\,\hat{a}[\hat{a}, \hat{a}^{\dagger}] = \hbar\omega\hat{a}$$

$$[\hat{a}^{\dagger}, \hat{H}]$$

$$= \hbar\omega\left[\hat{a}^{\dagger}, \hat{a}^{\dagger}\hat{a} + \frac{1}{2}\right] = \hbar\omega\,\hat{a}^{\dagger}[\hat{a}^{\dagger}, \hat{a}] = -\hbar\omega\hat{a}^{\dagger}$$

求$\hat{H}$的本征态（设为$|\lambda\rangle$）

$\hat{H}|\lambda\rangle = \lambda\hbar\omega|\lambda\rangle$

$\Rightarrow \langle\lambda|\hat{H}|\lambda\rangle = \hbar\omega\langle\lambda|\hat{a}^{+}\hat{a}+\frac{1}{2}|\lambda\rangle = \lambda\hbar\omega\langle\lambda|\lambda\rangle$

$0 \leq |a|\lambda\rangle|^2 = \langle\lambda|a^{+}a|\lambda\rangle = (\lambda-\frac{1}{2})\hbar\omega \rightarrow \lambda \geq \frac{1}{2}$

$$\begin{cases} \hat{H}\hat{a}|\lambda\rangle = (\hat{a}\hat{H} - \hbar\omega\hat{a})|\lambda\rangle = (\lambda-1)\hbar\omega\hat{a}|\lambda\rangle \\ \hat{H}\hat{a}^{+}|\lambda\rangle = (\hat{a}^{+}\hat{H} + \hbar\omega\hat{a}^{+})|\lambda\rangle = (\lambda+1)\hbar\omega\hat{a}|\lambda\rangle \end{cases}$$

$\Rightarrow$ $\hat{a}|\lambda\rangle$，$\hat{a}^{+}|\lambda\rangle$ 也是$\hat{H}$的本征态.（$\hat{a}|\lambda\rangle$，$\hat{a}^{+}|\lambda\rangle \neq 0$）

$\exists\ |\lambda_{min}\rangle$ s.t. $\hat{a}|\lambda_{min}\rangle = 0$（不存在$|\lambda_{max}\rangle$ s.t. $\hat{a}^{+}|\lambda\rangle \neq 0$）

本征值序列 $\hbar\omega\{\lambda, \lambda-1 \cdots\cdots \frac{1}{2}\}$

$E_n = (n+\frac{1}{2})\hbar\omega$，$\hat{a}^{+}$上升算符，$\hat{a}$下降算符.

记$\hat{H}$特征值 $n+\frac{1}{2}$ 对应的本征态为 $|n\rangle$

设 $\hat{a}|n\rangle = c|n-1\rangle$，求$c$的值.

归一化：$\langle n|\hat{a}^{+}\hat{a}|n\rangle = c^2\langle n-1|n-1\rangle = c^2$

$\langle n|\hat{H}|n\rangle = \langle n|\hbar\omega(\hat{a}^{+}\hat{a}+\frac{1}{2})|n\rangle = (n+\frac{1}{2})\hbar\omega$

$\Rightarrow c^2 = n$

$\hat{a}|n\rangle = \sqrt{n}|n-1\rangle$.

设 $\hat{a}^{+}|n\rangle = c'|n+1\rangle$，求$c'$的值.

$\langle n|\hat{a}\hat{a}^{+}|n\rangle = c'^2\langle n+1|n+1\rangle = c'^2$

$\langle n|\hat{H}|n\rangle = \hbar\omega\langle n|\hat{a}^{+}\hat{a}+\frac{1}{2}|n\rangle = \hbar\omega\langle n|\hat{a}\hat{a}^{+}-\frac{1}{2}|n\rangle$

$= (n+\frac{1}{2})\hbar\omega$

$\Rightarrow c'^2 = n+1$，$c' = \sqrt{n+1}$

$\hat{a}^{+}|n\rangle = \sqrt{n+1}|n+1\rangle$

$\Rightarrow \hat{a}^{+}\hat{a}|n\rangle = \hat{a}^{+}\sqrt{n}|n-1\rangle = n|n\rangle$ $|n\rangle$是数算符 $\hat{N} = \hat{a}^{+}\hat{a}$ 的本征态，本征值为n

$(\hat{a}^{+})^n|0\rangle = \sqrt{n!}|n\rangle$，$|n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^{+})^n|0\rangle$

在坐标表象下：H的本征态的形式

$\hat{a}|0\rangle = 0$,

$\langle x|\hat{a}|0\rangle = \frac{1}{\sqrt{2m\hbar\omega}}\langle x|i\hat{p}+m\omega\hat{x}|0\rangle = 0$

$\beta = \sqrt{\frac{m\omega}{\hbar}}$

$= \frac{1}{\sqrt{2m\hbar\omega}}(\hbar\frac{\partial}{\partial x}+m\omega x)\langle x|0\rangle = \sqrt{\frac{\hbar}{2m\omega}}(\frac{\partial}{\partial x}+\beta x)\langle x|0\rangle$

解得 $\langle x|0\rangle = Ce^{-\frac{1}{2}\beta^2x^2}$ 本征态 $|0\rangle$ 在 x 表象下的形式.

归一化： $\int_{-\infty}^{\infty}|\langle x|0\rangle|^2dx = C^2\int_{-\infty}^{\infty}e^{-\beta^2x^2}dx$

$= C^2\sqrt{\frac{\pi}{\beta^2}} = 1 \rightarrow C = \sqrt{\frac{\beta}{\sqrt{\pi}}}$

$\psi_0(x) = \sqrt{\frac{\beta}{\sqrt{\pi}}}e^{-\frac{1}{2}\beta^2x^2}$

$\langle\hat{x}|\hat{a}^+|\hat{x}\rangle = \langle\hat{x}|\frac{1}{\sqrt{2m\hbar\omega}}(-i\hat{p}+m\omega\hat{x})|\hat{x}\rangle$

$= \frac{1}{\sqrt{2m\hbar\omega}}(-\hbar\frac{\partial}{\partial x}+m\omega x) = \underline{\frac{1}{\sqrt{2}}(-\frac{1}{\beta}\frac{\partial}{\partial x}+\beta x)}$

$\langle x|\hat{a}|x\rangle = K$

$\Updownarrow$

$\langle x|\hat{a}^n|x\rangle = K^n$

$\langle x|n\rangle = \langle x|\frac{1}{\sqrt{n!}}(\hat{a}^+)^n|0\rangle$

$= \langle x|\frac{1}{\sqrt{n!}}(\hat{a}^+)^n|x\rangle\langle x|0\rangle$

$= \frac{1}{\sqrt{n!}}\underline{(\frac{1}{\sqrt{2}})^n(-\frac{1}{\beta}\frac{d}{dx}+\beta x)^n}\langle x|0\rangle$

$= (\frac{\beta}{\sqrt{\pi}})^{1/2}\frac{1}{\sqrt{2^nn!}}\underline{H_n(\beta x)}e^{-\frac{1}{2}\beta^2x^2}$ $H_n(x) = (-\frac{d}{dx}+x)^ne^{-\frac{1}{2}x^2}$

例：在 t=0 时刻，一个谐振子处的状态为 $|\psi\rangle = |1\rangle - i|2\rangle$

求坐标的不确定度.

$E_n = (n+\frac{1}{2})\hbar\omega$

$|\psi(t)\rangle = \frac{1}{\sqrt{2}}|1\rangle e^{-i\frac{3}{2}\omega t} - \frac{1}{\sqrt{2}}i|2\rangle e^{-i\frac{5}{2}\omega t}$

$\langle\psi^*(t)|\hat{x}|\psi(t)\rangle = \langle\psi^*(t)|\sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^++\hat{a})|\psi(t)\rangle$

$= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\Big((\langle1|e^{i\frac{3}{2}\omega t}+i\langle2|e^{i\frac{5}{2}\omega t})\hat{a}^+(|1\rangle e^{-i\frac{3}{2}\omega t}-i|2\rangle e^{-i\frac{5}{2}\omega t})$

$+(\langle1|e^{i\frac{3}{2}\omega t}+i\langle2|e^{i\frac{5}{2}\omega t})\hat{a}(|1\rangle e^{-i\frac{3}{2}\omega t}-i|2\rangle e^{-i\frac{5}{2}\omega t})\Big)$

$= \frac{1}{2}\sqrt{\frac{\hbar}{2m\omega}}\Big((\langle1|e^{i\frac{3}{2}\omega t}+i\langle2|e^{i\frac{5}{2}\omega t})(\sqrt{2}|2\rangle e^{-i\frac{3}{2}\omega t}-i\sqrt{3}|3\rangle e^{-i\frac{5}{2}\omega t})$

$+(\langle1|e^{i\frac{3}{2}\omega t}+i\langle2|e^{i\frac{5}{2}\omega t})(|0\rangle e^{-i\frac{3}{2}\omega t}-\sqrt{2}i|1\rangle e^{-i\frac{5}{2}\omega t})\Big)$

$= \frac{1}{\sqrt{2}}\sqrt{\frac{\hbar}{2m\omega}}(ie^{i\omega t}-ie^{-i\omega t}) = \sqrt{2}\sqrt{\frac{\hbar}{2m\omega}}\sin\omega t$

$$\langle \psi^*(t) | \hat{x}^2 | \psi(t) \rangle$$

$$= \langle \psi^*(t) | \frac{\hbar}{2m\omega} \left(\hat{a}^{+2} + \hat{a}^{+}\hat{a} + \hat{a}\hat{a}^{+} + \hat{a}^{2} \right) | \psi(t) \rangle$$

$$= \frac{\hbar}{4m\omega} \Big(\big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}^{+2} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big)$$

$$+ \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}^{+}\hat{a} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big)$$

$$+ \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}\hat{a}^{+} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big)$$

$$+ \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}^{2} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big) \Big)$$

$$= \frac{\hbar}{4m\omega} \Big(\big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \big(\sqrt{6}|3\rangle e^{-i\frac{3}{2}\omega t} - i\sqrt{12}|4\rangle e^{-i\frac{5}{2}\omega t}\big)$$

$$+ \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}^{+}\hat{a} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big)$$

$$+ \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}\hat{a}^{+} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big) \qquad \hat{a}\hat{a}^{+} = \hat{a}^{+}\hat{a} + 1$$

$$+ \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \big(-i|0\rangle e^{-i\frac{5}{2}\omega t}\big) \Big) \qquad = \hat{N} + 1$$

$$= \frac{\hbar}{4m\omega} \Big[2\cdot \big(\langle 1| e^{i\frac{3}{2}\omega t} + i\langle 2| e^{i\frac{5}{2}\omega t}\big) \hat{a}^{+}\hat{a} \big(|1\rangle e^{-i\frac{3}{2}\omega t} - i|2\rangle e^{-i\frac{5}{2}\omega t}\big)$$

$$+ (1+1) \Big]$$

$$= \frac{\hbar}{4m\omega} \Big[2\cdot(1+2) + 2 \Big] = \frac{\hbar}{2m\omega} \cdot 4$$

$$\langle \Delta x \rangle = \sqrt{\langle \hat{x}^2 \rangle - \langle \hat{x} \rangle^2}$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \sqrt{4 - 2\sin^2 \omega t}$$

$$= \sqrt{\frac{\hbar}{m\omega}} \sqrt{2 - \sin^2 \omega t}$$

§ 3.2 角动量

$\hat{L} = \hat{r} \times \hat{p}$

$[\hat{L}_i, \hat{L}_j] = \varepsilon_{ijk}\, i\hbar L_k e_k$

$\hat{L}^2 = \hat{L} \cdot \hat{L} = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$

$[\hat{L}^2, \hat{L}_i] = 0$

⇒ 选取 $\{\hat{L}^2, \hat{L}_z\}$ 为力学量完全集

共同本征态，设为 $|\lambda\rangle$

定义 $\hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y$

$$\begin{cases} \hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-) \\ \hat{L}_y = \frac{1}{2i}(\hat{L}_+ - \hat{L}_-) \end{cases}$$

$[\hat{L}_z, \hat{L}_\pm] = [\hat{L}_z, \hat{L}_x] \pm i[\hat{L}_z, \hat{L}_y]$

$= i\hbar(L_y \mp iL_x)$

$= \pm\hbar\hat{L}_\pm$

$[\hat{L}^2, \hat{L}_\pm] = [\hat{L}^2, L_x] \pm i[\hat{L}^2, \hat{L}_y] = 0$

$\hat{L}^2(L_\pm|\lambda\rangle) = \hat{L}_\pm(\hat{L}^2|\lambda\rangle) = \mu\hat{L}_\pm|\lambda\rangle$

$\hat{L}_z(\hat{L}_\pm|\lambda\rangle) = \pm\hbar\hat{L}_\pm|\lambda\rangle + L_\pm L_z|\lambda\rangle = (\nu \pm \hbar)(\hat{L}_\pm|\lambda\rangle)$

$[L_z, L_\pm] = \hat{L}_z\hat{L}_\pm - \hat{L}_\pm\hat{L}_z = \pm\hbar\hat{L}_\pm$

设 $l\hbar$ 为 L_z 的最大本征值，对应本征态为 $|\lambda_+\rangle$

此时 $\hat{L}_+|\lambda_+\rangle = 0$

$\hat{L}_z|\lambda\rangle = l\hbar|\lambda\rangle$，$\hat{L}^2|\lambda_+\rangle = \mu|\lambda_+\rangle$

$\hat{L}^2 = \hat{L}_\pm\hat{L}_\mp + \hat{L}_z^2 \mp \hbar\hat{L}_z$

$\hat{L}^2|\lambda_+\rangle = \hat{L}_-\hat{L}_+|\lambda_+\rangle + \hat{L}_z^2|\lambda_+\rangle + \hbar\hat{L}_z|\lambda_+\rangle$

$= l^2\hbar^2|\lambda_+\rangle + l\hbar^2|\lambda_+\rangle$

$= l(l+1)\hbar^2|\lambda_+\rangle$

同理：存在 $\bar{l}\hbar$ 为 $\hat{L}_z$ 的最小本征值，对应本征态为 $|\lambda_0\rangle$

此时 $\hat{L}$ $|\lambda\rangle =$

$\hat{L}_z|\lambda_b\rangle = \bar{l}\hbar|\lambda_b\rangle$, $\hat{L}^2|\lambda_b\rangle = \mu|\lambda_b\rangle$

$\hat{L}^2|\lambda_b\rangle = \hat{L}_+\hat{L}_-|\lambda_b\rangle + \hat{L}_z^2|\lambda_b\rangle - \hbar\hat{L}_z|\lambda_b\rangle$

$= \bar{l}(\bar{l}-1)\hbar^2|\lambda_b\rangle$

$|\lambda_t\rangle$, $|\lambda_b\rangle$ 都是 $\hat{L}^2$ 的本征态.

$\Rightarrow$ $L^2 = l(l+1) = \bar{l}(\bar{l}-1)$

解得 $\bar{l} = -l$ 或 $l+1$ (舍去)

由 $\hat{L}_\pm$ 性质: $l - \bar{l} = 2l = N$ (整数)

$\Rightarrow$ 共同本征态 $|l,m\rangle$ $(m = -l, -l+1, \cdots, l)$

$\hat{L}^2|l,m\rangle = l(l+1)\hbar^2|l,m\rangle$. $\hat{L}_z|l,m\rangle = m\hbar|l,m\rangle$

升降算符系数:

$\hat{L}_-|l,m\rangle = C_-\hbar|l,m-1\rangle$

$\langle l,m|\hat{L}_+\hat{L}_-|l,m\rangle = |C_-|^2\hbar^2\langle l,m-1|l,m-1\rangle$

$\hat{L}_+\hat{L}_- = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z$

$= \langle l,m|\hat{L}^2|l,m\rangle - \langle l,m|\hat{L}_z^2|l,m\rangle + \hbar\langle l,m|\hat{L}_z|l,m\rangle$

$= l(l+1)\hbar^2 - m^2\hbar^2 - m\hbar = [l(l+1) - m(m-1)]\hbar$

$\Rightarrow$ $C_- = \sqrt{l(l+1) - m(m-1)}$

$\hat{L}_-|l,m\rangle = \sqrt{l(l+1) - m(m-1)}$

设 $\hat{L}_+|l,m\rangle = C_+\hbar|l,m+1\rangle$

$\langle l,m|\hat{L}_-\hat{L}_+|l,m\rangle$ $\quad$ $\hat{L}_-\hat{L}_+ = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z$

$= [l(l+1) - m^2 - m]\hbar^2$

$\Rightarrow$ $C_+ = \sqrt{l(l+1) - m(m+1)}$

$\hat{L}_+|l,m\rangle = \sqrt{l(l+1) - m(m+1)}\,\hbar\,|l,m+1\rangle$

$L_\pm|l,m\rangle = C_\pm\hbar|l,m\pm1\rangle$

$C_\pm = \sqrt{l(l+1) - m(m\pm1)}$

轨道角动量在球坐标下的具体形式：

$Y_{lm}(\theta,\varphi) = \langle\theta,\varphi|l,m\rangle$

$\hat{\vec{L}} = \hat{\vec{r}} \times \hat{\vec{p}}$

$$\begin{cases} \hat L_x = -i\hbar\left(-\sin\phi\frac{\partial}{\partial\theta} - \cos\phi\cot\theta\frac{\partial}{\partial\phi}\right) \\ \hat L_y = -i\hbar\left(\cos\phi\frac{\partial}{\partial\theta} - \sin\phi\cot\theta\frac{\partial}{\partial\phi}\right) \\ \hat L_z = -i\hbar\frac{\partial}{\partial\phi} \end{cases}$$

$$\hat L_\pm = \hat L_x \pm i\hat L_y = -i\hbar\left[(-\sin\phi \pm i\cos\phi)\frac{\partial}{\partial\theta} + (-\cos\phi \mp i\sin\phi)\cot\theta\frac{\partial}{\partial\phi}\right]$$

$$= i\hbar\left[(\sin\phi \mp i\cos\phi)\frac{\partial}{\partial\theta} + (\cos\phi \pm i\sin\phi)\cot\theta\frac{\partial}{\partial\phi}\right]$$

$$= \hbar\left[(\pm\cos\phi + i\sin\phi)\frac{\partial}{\partial\theta} + i e^{\pm i\phi}\cot\theta\frac{\partial}{\partial\phi}\right]$$

$$= \hbar e^{\pm i\phi}\left(\pm\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right) \qquad + = \pm \cdot \pm$$

$$= \pm\hbar e^{\pm i\phi}\left(\frac{\partial}{\partial\theta} \pm i\cot\theta\frac{\partial}{\partial\phi}\right)$$

$$\hat L^2 = \hat L_x^2 + \hat L_y^2 + \hat L_z^2$$

$$= -\hbar^2\Big[\sin^2\phi\frac{\partial^2}{\partial\theta^2} + \sin\phi\cos\phi\frac{\partial}{\partial\theta}\left(\cot\theta\frac{\partial}{\partial\phi}\right) + \cos^2\phi\cot\theta\frac{\partial^2}{\partial\phi^2} + \cos\phi\cot\theta\frac{\partial}{\partial\phi}\left(\sin\phi\frac{\partial}{\partial\theta}\right)$$

$$+ \cos^2\phi\frac{\partial^2}{\partial\theta^2} - \cos\phi\sin\phi\frac{\partial}{\partial\theta}\left(\cot\theta\frac{\partial}{\partial\phi}\right) + \sin^2\phi\cot^2\theta\frac{\partial^2}{\partial\phi^2} - \sin\phi\cot\theta\frac{\partial}{\partial\phi}\left(\cos\phi\frac{\partial}{\partial\theta}\right) + \frac{\partial^2}{\partial\phi^2}$$

$$= -\hbar^2\left[\frac{\partial^2}{\partial\theta^2} + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2} + \cot\theta(\cos^2\phi + \sin^2\phi)\frac{\partial}{\partial\theta}\right]$$

$$= -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]$$

由 $\hat L_+|l,l\rangle = 0$ $\qquad \hat L_+|\lambda_+\rangle = 0$

$$\hbar e^{i\phi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right)Y_{ll}(\theta,\varphi) = 0$$

分离变量法 $Y_{ll}(\theta,\varphi) = f(\theta)g(\varphi)$ $\qquad \hat L_z = -i\hbar\frac{\partial}{\partial\varphi}$

以 $g\frac{\partial f}{\partial\theta} + i\cot\theta f\frac{\partial g}{\partial\phi} = 0$， $\frac{f'}{f}\tan\theta = -i\frac{g'}{g} = C$ $\qquad \hat L_z Y_{ll}(\theta,\varphi) = l\hbar fg$

$\frac{f'}{f} = l\cot\theta$， $\frac{g'}{g} = il$ $\qquad = -i\hbar f g' \rightarrow C = l$

解得 $f(\theta) = e^{C'}\sin^l\theta$ ， $g(\varphi) = e^{i(l+C_2)\varphi}$

$Y_{ll}(\theta,\varphi) \propto \sin^l\theta\, e^{il\varphi}$

应用下降算符： $Y_{lm}(\theta,\varphi) = \langle\theta,\varphi|l,m\rangle = \langle\theta,\varphi|\hat L_-^{l-m}|l,l\rangle$

$$= \alpha_m (\hat{L}_-)^{l-m} \sin^l\theta \, e^{il\varphi}$$

归一化的本征态 球谐函数 $Y_{lm}(\theta,\varphi) = (-1)^m \sqrt{\frac{2l+1}{4\pi}\frac{(l-m)!}{(l+m)!}} P_l^m(\cos\theta)\, e^{im\varphi}$

例：在 $(\hat{L}^2, \hat{L}_z)$ 表象下，粒子运动状态的角度部分可用如下函数描写.

$$\psi(\theta,\varphi) = Y_{00}(\theta,\varphi) + Y_{10}(\theta,\varphi) + \sqrt{2}\, Y_{11}(\theta,\varphi)$$

(1) 求 $\hat{L}^2$ 的期待值.

归一化：$\psi^*\psi = 1+1+2 = 4 \rightarrow \varphi = \frac{1}{2}\psi$

$$\langle \psi^* | \hat{L}^2 | \psi \rangle = \frac{1}{4}(0 + 1\cdot 2 + 2\cdot 1\cdot 2)\hbar^2 = \frac{3}{2}\hbar^2$$

(2) 求 $L_n = \vec{L}\cdot\vec{n}$，$\vec{n} = (\sin\alpha\cos\beta, \sin\alpha\sin\beta, \cos\beta)$

的可能测量值和概率

$l = 0$：0. $P = |\langle 0,0|\psi\rangle|^2 = \frac{1}{4}$

$l = 1$：$|1,1\rangle$ $|1,0\rangle$，$|1,-1\rangle$ 为基底.

此时 $\hat{L}_+|1,1\rangle = 0$，$\hat{L}_+|1,0\rangle = \sqrt{1(1+1) - 0(0+1)}\hbar|1,1\rangle = \sqrt{2}\hbar|1,1\rangle$

$\hat{L}_+|1,-1\rangle = \sqrt{1\cdot(1+1) - (-1)(-1+1)}\hbar|1,0\rangle = \sqrt{2}\hbar|1,0\rangle$

$$\hat{L}_+ : \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}\hbar$$

$\hat{L}_-|1,1\rangle = \sqrt{1(1+1) - 1(1-1)}\hbar|1,0\rangle = \sqrt{2}\hbar|1,0\rangle$

$\hat{L}_-|1,0\rangle = \sqrt{1(1+1) - 0(0-1)}\hbar|1,-1\rangle = \sqrt{2}\hbar|1,-1\rangle$，$\hat{L}_-|1,-1\rangle = 0$

$$\hat{L}_- : \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}\hbar$$

$$\hat{L}_x = \frac{1}{2}(\hat{L}_+ + \hat{L}_-) = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}\hbar$$

$$\hat{L}_y = \frac{1}{2i}(\hat{L}_+ - \hat{L}_-) = \frac{1}{\sqrt{2}i}\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}\hbar$$

$$\hat{L}_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}\hbar$$

$$L_n = (\hat{L}_x, \hat{L}_y, \hat{L}_z)\cdot(\sin\alpha\cos\beta, \sin\alpha\sin\beta, \cos\alpha)$$

$$= \frac{\hbar}{\sqrt{2}}\begin{pmatrix} \sqrt{2}\cos\alpha & \sin\alpha\, e^{i\beta} & 0 \\ \sin\alpha\, e^{-i\beta} & 0 & \sin\alpha\, e^{i\beta} \\ 0 & \sin\alpha\, e^{-i\beta} & -\sqrt{2}\cos\alpha \end{pmatrix}$$

$$|L_n - \lambda\frac{\hbar}{\sqrt{2}} I|$$

$\hbar^3|$

$= (\frac{}{\sqrt{2}}) \quad (\sqrt{2}\cos\alpha - \lambda) - \lambda)(-\sqrt{2}\cos\alpha - \lambda) \qquad 2\hbar \quad -1 \quad - \sin \quad -2\cos$

$= (\frac{\hbar}{\sqrt{2}})^3 \left[(2\cos^2\alpha - \lambda^2)\lambda + 2\sin^2\alpha\,\lambda \right] = (\frac{\hbar}{\sqrt{2}})^3 \lambda(\lambda^2 - 2) = 0$

$\rightarrow \lambda = \pm\sqrt{2}, 0$， 对应本征值，$\pm\hbar$，$0$

① $\lambda = \sqrt{2}$

$$\begin{pmatrix} \sqrt{2}(\cos\alpha - 1) & \sin\alpha\, e^{i\beta} & 0 \\ \sin\alpha\, e^{-i\beta} & -\sqrt{2} & \sin\alpha\, e^{i\beta} \\ 0 & \sin\alpha\, e^{-i\beta} & -\sqrt{2}(\cos\alpha + 1) \end{pmatrix} |\lambda_1\rangle = 0$$

$$|\lambda_1\rangle = \begin{pmatrix} e^{i\beta}\cos^2\frac{\alpha}{2} \\ \frac{\sqrt{2}}{2}\sin\alpha \\ e^{-i\beta}\sin^2\frac{\alpha}{2} \end{pmatrix} \qquad P_1 = |\langle\psi|\lambda_1\rangle|^2 = |\frac{\sqrt{2}}{4}\sin\alpha + \frac{\sqrt{2}}{2}e^{i\beta}\cos^2\frac{\alpha}{2}|^2$$

$$= (\frac{\sqrt{2}}{4}\sin\alpha + \frac{\sqrt{2}}{2}\cos^2\frac{\alpha}{2}\cos\beta)^2 + (\frac{\sqrt{2}}{2}\cos^2\frac{\alpha}{2}\sin\beta)^2$$

② $\lambda = -\sqrt{2}$

$$|\lambda_2\rangle = \begin{pmatrix} -e^{i\beta}\sin^2\frac{\alpha}{2} \\ \frac{\sqrt{2}}{2}\sin\alpha \\ -e^{-i\beta}\cos^2\frac{\alpha}{2} \end{pmatrix} \qquad P_2 = |\langle\psi|\lambda_2\rangle|^2 = (\frac{\sqrt{2}}{4}\sin\alpha - \frac{\sqrt{2}}{2}e^{i\beta}\sin^2(\frac{\alpha}{2}))^2$$

$$= (\frac{\sqrt{2}}{4}\sin\alpha - \frac{\sqrt{2}}{2}\sin^2\frac{\alpha}{2}\cos\beta)^2 + (\frac{\sqrt{2}}{2}\sin^2\frac{\alpha}{2}\sin\beta)^2$$

③ $\lambda = 0$

$$|\lambda_3\rangle = \begin{pmatrix} -\frac{\sqrt{2}}{2}e^{i\beta}\sin\alpha \\ \cos\alpha \\ \frac{\sqrt{2}}{2}e^{-i\beta}\sin\alpha \end{pmatrix} \qquad P_3 = |\langle\psi|\lambda_3\rangle^2| = (\frac{1}{2}\cos\alpha - \frac{1}{2}e^{i\beta}\sin\alpha)^2$$

$$= (\frac{1}{2}\cos\alpha - \frac{1}{2}\sin\alpha\cos\beta)^2 + (\frac{1}{2}\sin\alpha\sin\beta)^2$$

$P(\lambda = \hbar) = P_1$，$P(\lambda = 0) = P_2 + \frac{1}{4}$，$P(\lambda = -\hbar) = P_3$

作业

1. (1) $\langle x|\psi_0\rangle = \left(\frac{m\omega}{\hbar\pi}\right)^{1/4} e^{-\frac{1}{2}\frac{m\omega x^2}{\hbar}}$

$$\psi(p) = \langle p|\psi_0\rangle = \langle p|x\rangle\langle x|\psi_0\rangle$$

$$= \frac{1}{\sqrt{2\pi\hbar}}\int dx\, e^{-\frac{ipx}{\hbar}}\left(\frac{m\omega}{\hbar\pi}\right)^{1/4} e^{-\frac{1}{2}\frac{m\omega x^2}{\hbar}}$$

$$= \frac{1}{\sqrt{2\pi\hbar}}\left(\frac{m\omega}{\hbar\pi}\right)^{1/4}\int dx\, e^{-\frac{m\omega}{2\hbar}x^2 - \frac{ip}{\hbar}x}$$

$T = \frac{p^2}{2m}$, $P(T)\,dT = P(p)\,dp$, $P(p) = |\psi_0(p)|^2$

$$P(T) = \frac{dp}{dT}\, dP(p) = \frac{m}{p}\, dP(p)$$

(2) $C(t) = \langle 0| e^{\frac{i\hat{H}t}{\hbar}}\hat{x}\, e^{-\frac{i\hat{H}t}{\hbar}}\hat{x}|0\rangle \qquad x \to \sqrt{\frac{\hbar}{2m\omega}}(\hat{a}^\dagger + \hat{a})$

$$= \frac{\hbar}{2m\omega}\langle 0| e^{\frac{i\hat{H}t}{\hbar}}(\hat{a}^\dagger + \hat{a})\, e^{-\frac{i\hat{H}t}{\hbar}}(\hat{a}^\dagger + \hat{a})|0\rangle$$

$$= \frac{\hbar}{2m\omega} e^{\frac{iE_0 t}{\hbar}} e^{-\frac{iE_1 t}{\hbar}} = \frac{\hbar}{2m\omega} e^{-i\omega t}$$

2. (1) $|\alpha\rangle = \sum_i \alpha_i |i\rangle$

$$\hat{a}|\alpha\rangle = \sum_i \alpha_i \hat{a}|i\rangle = \sum_i \alpha_i\sqrt{i}\,|i-1\rangle = \alpha|\alpha\rangle = \sum_i \alpha_i\alpha\,|i\rangle$$

$\alpha_0 = 0$, $\alpha_1 = 1$, $\alpha_{n+1}\sqrt{n+1} = \alpha\alpha_n$

$$|\alpha\rangle = \sum_i \frac{\alpha^i}{\sqrt{i!}}|i\rangle$$

(2) $|\alpha(t)\rangle = \sum_i \frac{\alpha^i}{\sqrt{i!}} e^{-\frac{i\hbar\omega(i+\frac{1}{2})t}{\hbar}} = \sum_i \frac{\alpha^i}{\sqrt{i!}} e^{-i(i+\frac{1}{2})\omega t}|i\rangle$

$$\langle x\rangle = \langle\alpha(t)|\hat{x}|\alpha(t)\rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\sum_i \frac{\alpha^i}{\sqrt{i!}} e^{i(i+\frac{1}{2})\omega t}\langle i|(\hat{a}^\dagger + \hat{a})\sum_k \frac{\alpha^k}{\sqrt{k!}} e^{-i(k+\frac{1}{2})\omega t}|k\rangle$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\sum_i \frac{\alpha^i}{\sqrt{i!}}\left|\frac{\alpha^{i-1}}{\sqrt{(i-1)!}}\cdot\sqrt{i}\, e^{i\omega t} + \frac{\alpha^{i+1}}{\sqrt{(i+1)!}}\cdot\sqrt{i+1}\, e^{-i\omega t}\right|$$

$$= \sqrt{\frac{\hbar}{2m\omega}} \sum_i \left[\frac{\alpha}{(i+1)!} e^{\omega t} + \frac{\alpha}{i!} e^{-i\omega t}\right.$$

(3) 令 $\hat{b} = \hat{a} + \frac{V}{\hbar\omega}$

$$\hat{H} = \hbar\omega \hat{a}^+ \hat{a} + V^* \hat{a} + V \hat{a}^+$$

$$= \hbar\omega \hat{b}^+ \hat{b} - \frac{V^* V}{\hbar\omega}$$

$$\hat{H} |\lambda\rangle = \lambda \hbar\omega |\lambda\rangle$$

仿照升降算符.

$$[b, \hat{b}^+] = (\hat{b}\hat{b}^+ - \hat{b}^+\hat{b}) = \hat{a}\hat{a}^+ - \hat{a}^+\hat{a} = n+1-n = 1$$

$$[\hat{b}, \hat{H}] = \hbar\omega [\hat{b}, \hat{b}^+\hat{b}] - \frac{V^*V}{\hbar\omega}\hat{b} = (\hbar\omega[\hat{b}, \hat{b}^+] - \frac{V^*V}{\hbar\omega})\hat{b}$$

$$= (\hbar\omega - \frac{V^*V}{\hbar\omega})\hat{b}$$

$$[\hat{b}^+, \hat{H}] = \hbar\omega [\hat{b}^+, \hat{b}^+\hat{b}] - \frac{V^*V}{\hbar\omega}\hat{b}^+ = (\hbar\omega \hat{b}^+ [\hat{b}^+, \hat{b}] - \frac{V^*V}{\hbar\omega}\hat{b}^+)$$

$$= -(\hbar\omega + \frac{V^*V}{\hbar\omega})\hat{b}^+$$

$$\hat{H}\hat{b}|\lambda\rangle = (\hat{b}\hat{H} - [\hat{b}, \hat{H}])|\lambda\rangle$$

$$= \hat{b}(\hbar\omega\lambda - \hbar\omega + \frac{V^*V}{\hbar\omega})|\lambda\rangle$$

$$= \left[\hbar\omega(\lambda-1) + \frac{V^*V}{\hbar\omega}\right]\hat{b}|\lambda\rangle$$

$$\hat{H}\hat{b}^+|\lambda\rangle = (\hat{b}^+\hat{H} - [\hat{b}^+, \hat{H}])|\lambda\rangle$$

$$= \hat{b}^+(\hbar\omega\lambda + \hbar\omega + \frac{V^*V}{\hbar\omega})|\lambda\rangle$$

$$= \left[\hbar\omega(\lambda+1) + \frac{V^*V}{\hbar\omega}\right]\hat{b}^+|\lambda\rangle$$

本征值 $E_n = n\hbar\omega + \frac{V^*V}{\hbar\omega}$

$$\hat{b}|\lambda_0\rangle = 0 \rightarrow \hat{a}|\lambda_0\rangle = -\frac{V}{\hbar\omega}|\lambda_0\rangle$$

为(1)中 $\alpha = -\frac{V}{\hbar\omega}$ 的情形,

3. $\hat{H}|l,m\rangle = \frac{1}{2I} l(l+1)\hbar^2 + \frac{4}{3I} m\hbar^2$

$= \frac{\hbar^2}{6I}\left[3l(l+1) + 8m\right]$

$\geq \frac{\hbar^2}{6I}(3l^2 - 5l) \geq \frac{\hbar^2}{6I}(-2) = -\frac{\hbar^2}{3I}$ (此时 $l=1$, $m=-l=-1$)

在 3 空间 $\{|1,1\rangle, |1,0\rangle, |1,-1\rangle\}$ 中.

$\hat{H} = \frac{\hat{L}^2}{2I} + \frac{4\hbar}{3I}\frac{1}{2i}(\hat{L}_+ - \hat{L}_-)$

$= \frac{\hbar^2}{2I}\begin{pmatrix} 2 & & 0 \\ & 2 & \\ 0 & & 2 \end{pmatrix} - i\frac{2\hbar^2}{3I}\left[\begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} - \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}\right]$

$= \frac{\hbar^2}{I}\begin{pmatrix} 1 & \beta i & 0 \\ -\beta i & 1 & \beta i \\ 0 & -\beta i & 1 \end{pmatrix}$

$|\hat{H} - \lambda I| =$

$|\psi_0\rangle = |1,-1\rangle = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

$|\psi(t)\rangle =$

$\langle \hat{L}_z \rangle = \langle \psi^*(t) | \hat{L}_z | \psi(t) \rangle$